PostgreSQL Operators – below version for 15

Operators in the WHERE clause

We can operate with different operators in the WHERE clause:

=	Equal to
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
<>	Not equal to
!=	Not equal to
LIKE	Check if a value matches a pattern (case sensitive)
ILIKE	Check if a value matches a pattern (case insensitive)
AND	Logical AND
OR	Logical OR
IN	Check if a value matches any value within a provided list
BETWEEN	Check if a value is within a specified range
IS NULL	Check if a value is NULL
NOT	Makes a negative result e.g. NOT LIKE, NOT IN, NOT BETWEEN

Equal To

The = operator is used when you want to return all records where a column is equal to a specified value:

Example

Return all records where the brand is 'Volvo':

```
SELECT * FROM cars
WHERE brand = 'Volvo';
```

Less Than

The < operator is used when you want to return all records where a column is less than a specified value.

Example

Return all records where the year is less than 1975:

```
SELECT * FROM cars
WHERE year < 1975;
```

Greater Than

The > operator is used when you want to return all records where a columns is greater than a specified value.

Example

Return all records where the year is greater than 1975:

```
SELECT * FROM cars
WHERE year > 1975;
```

Less Than or Equal To

The <= operator is used when you want to return all records where a column is less than, or equal to, a specified value.

Example

Return all records where the year is less than or equal to 1975:

```
SELECT * FROM cars
WHERE year <= 1975;
```

Greater Than or Equal to

The >= operator is used when you want to return all records where a columns is greater than, or equal to, a specified value.

Example

Return all records where the year is greater than or equal 1975:

```
SELECT * FROM cars
WHERE year >= 1975;
```

Not Equal To

The <> operator is used when you want to return all records where a column is NOT equal to a specified value:

Example

Return all records where the brand is NOT 'Volvo':

```
SELECT * FROM cars
WHERE brand <> 'Volvo';
```

You will get the same result with the != operator:

Example

Return all records where the brand is NOT 'Volvo':

```
SELECT * FROM cars
WHERE brand != 'Volvo';
```

LIKE

The LIKE operator is used when you want to return all records where a column is equal to a specified pattern.

The pattern can be an absolute value like 'Volvo', or with a wildcard that has a special meaning.

There are two wildcards often used in conjunction with the LIKE operator:

- The percent sign %, represents zero, one, or multiple characters.
- The underscore sign _, represents one single character.

Example

Return all records where the model STARTS with a capital 'M':

```
SELECT * FROM cars
WHERE model LIKE 'M%';
```

The LIKE operator is case sensitive.

ILIKE

Same as the LIKE operator, but ILIKE is case insensitive.

Example

Return all records where the model start with a 'm':

```
SELECT * FROM cars
WHERE model ILIKE 'm%';
```

AND

The logical AND operator is used when you want to check more that one condition:

Example

Return all records where the brand is 'Volvo' and the year is 1968:

```
SELECT * FROM cars
WHERE brand = 'Volvo' AND year = 1968;
```

OR

The logical OR operator is used when you can accept that only one of many conditions is true:

Example

Return all records where the brand is 'Volvo' OR the year is 1975:

```
SELECT * FROM cars
WHERE brand = 'Volvo' OR year = 1975;
```

IN

The IN operator is used when a column's value matches any of the values in a list:

Example

Return all records where the brand is present in this list: ('Volvo', 'Mercedes', 'Ford'):

```
SELECT * FROM cars
WHERE brand IN ('Volvo', 'Mercedes', 'Ford');
```

BETWEEN

The BETWEEN operator is used to check if a column's value is between a specified range of values:

Example

Return all records where the year is between 1970 and 1980:

```
SELECT * FROM cars
WHERE year BETWEEN 1970 AND 1980;
```

The BETWEEN operator includes the *from* and *to* values, meaning that in the above example, the result would include cars made in 1970 and 1980 as well.

IS NULL

The IS NULL operator is used to check if a column's value is NULL:

Example

Return all records where the model is NULL:

```
SELECT * FROM cars
WHERE model IS NULL;
```

NOT

The NOT operator can be used together with LIKE, ILIKE, IN, BETWEEN, and NULL operators to reverse the truth of the operator.

Example: NOT LIKE

Return all records where the brand does NOT start with a capital 'B' (case sensitive):

```
SELECT * FROM cars
WHERE brand NOT LIKE 'B%';
```

Example: NOT ILIKE

Return all records where the brand does NOT start with a 'b' (case insensitive):

```
SELECT * FROM cars
WHERE brand NOT ILIKE 'b%';
```

Example: NOT IN

Return all records where the brand is NOT present in this list: ('Volvo', 'Mercedes', 'Ford'):

```
SELECT * FROM cars
WHERE brand NOT IN ('Volvo', 'Mercedes', 'Ford');
```

Example: NOT BETWEEN

Return all records where the year is NOT between 1970 and 1980:

```
SELECT * FROM cars
WHERE year NOT BETWEEN 1970 AND 1980;
```

The NOT BETWEEN operator excludes the *from* and *to* values, meaning that in the above example, the result would not include cars made in 1970 and 1980.

Example: IS NOT NULL

Return all records where the model is NOT null:

```
SELECT * FROM cars
WHERE model IS NOT NULL;
```